Fault Lines of Inequality: When Earthquakes Discriminate

COM-480 Data Visualization - Milestone 2

Website: com-480-data-visualization.github.io/the-visualizers/

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1 Project Goal

In January 2010 a M 7.0 earthquake killed more than 222,000 people in Haiti. Seven weeks later a M 8.8 earthquake, releasing roughly 500× more energy, struck Chile and killed fewer than 600. Our project is a **scrollytelling interactive website** that uses this contrast to argue that earthquake mortality is driven by wealth, infrastructure, and preparedness rather than seismic force alone. We combine the NOAA Significant Earthquake Database (2,679 events, 1960-2026) with World Bank and UNDP indicators (GDP per capita, HDI, hospital beds, urbanisation, population density) to take the reader from a global overview down to country-level comparison.

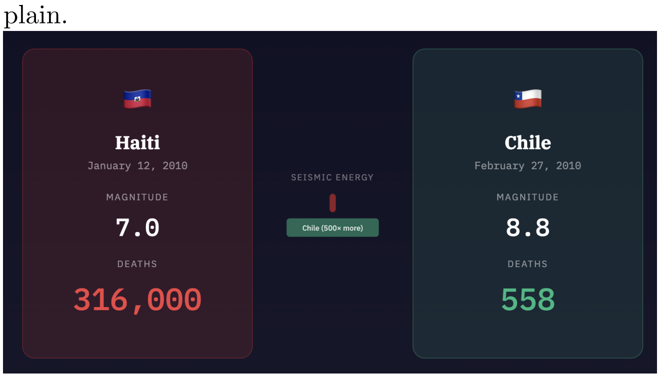
The site is organised as a single scrolling page with a sticky top navigation, a scroll-progress bar, and six sections: a hook, four scroll-driven visualisations, and an exploratory country-comparison tool.

2 Visualization Design and Sketches

Each section below is described with a sketch/screenshot, the intent, the tools and course lectures it draws on.

2.1 V1 - The Hook (Haiti vs. Chile)

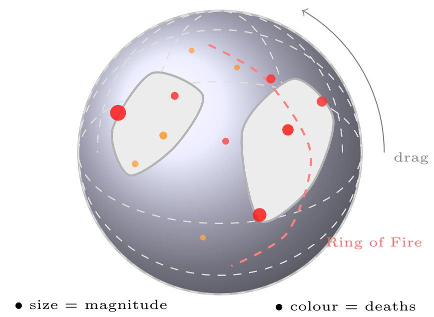
Two cards on a dark background show flag, magnitude, and a counter indicating the deaths. Between them, a horizontal bar encodes the seismic-energy ratio (Chile's fill is 500× wider). A short caption below asks how the smaller quake caused 396× more deaths, introducing the reader to the paradox that the rest of the site will explain.



Tools: HTML/CSS, D3 tween counters, IntersectionObserver. **Lectures:** *Designing Viz*; *JavaScript*; *D3.*

2.2 V2 - Interactive Globe

This is the core 3D mapping visualization showing earthquake locations, magnitudes, and fatalities with drag rotation and auto-rotation. Four scroll steps: (1) uniform dots, (2) radius $\propto$ magnitude, (3) recolour by death counter, (4) auto-rotate to frame the Pacific Ring of Fire. Dragging rotates the globe; hovering shows a tooltip with info about that earthquake.



Tools: D3 `geoOrthographic`, TopoJSON, D3 drag, `d3.timer`.

Lectures:

Maps: Understanding geographical projections and coordinate systems

Practical maps: Implementing D3 geoProjections, drawing paths, and using TopoJSON

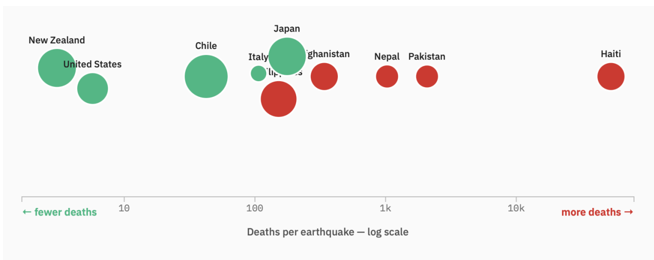
D3.js, Interactions and More interactive d3.js : Data binding, drawing SVG elements, implementing drag events and timers for auto-rotation, mouseover tooltips and drag behaviors

Perception colors: Using color scales and opacities to represent the number of deaths effectively

Mark, channels: Mapping data attributes like magnitude to circle radius and deaths to color

2.3 V3 - Magnitude Myth

One axis plot that shows that wealth explains deaths better than magnitude. Each circle represents an earthquake-prone country, positioned along a single axis by its average deaths per earthquake, with circle size encoding average magnitude and color indicating World Bank income group.



Tools: D3 `forceSimulation`

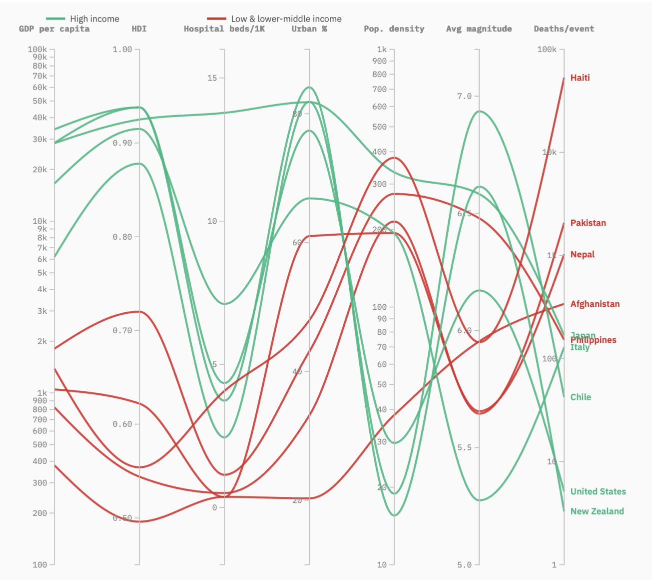
Lectures:

Tabular data: Visualizing distributions along a single axis

Interactions and More interactive d3.js: Tooltips and mouse events

2.4 V4 - Follow the Profile

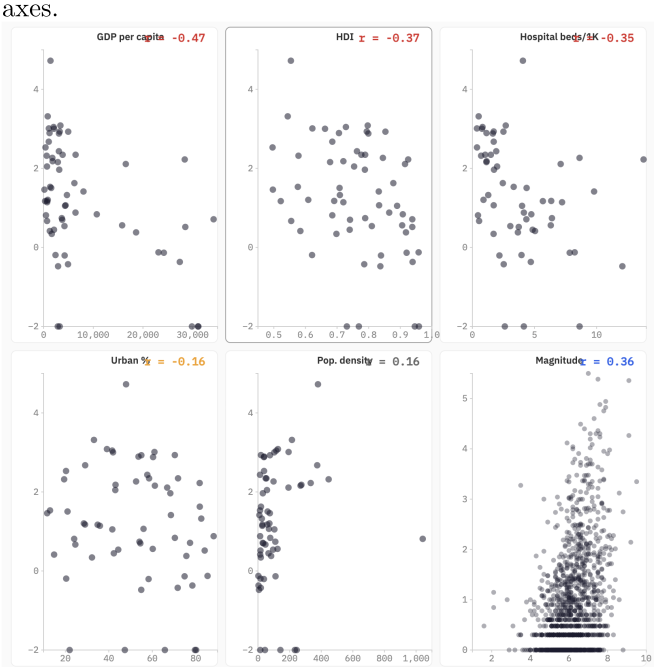
A parallel-coordinates plot linking the same countries across seven axes - GDP, HDI, hospital beds, urbanisation, population density, average magnitude, and deaths per event - where each line traces a country's profile end-to-end. Lines are colored by income group (green for high-income, red for low and lower-middle), revealing a clear separation: red trajectories cluster low on development indicators and rise sharply on deaths, while green lines sit high on development axes and terminate low on deaths, with hover revealing each country's full profile.



Tools: D3 `scalePoint`, `curveMonotoneX`
Lectures:
Tabular data: Visualizing highly multidimensional continuous data
D3.js: Generating SVG paths, mapping multiple domains
Perception colors: Categorical color palettes for income buckets

2.5 V5 - Correlation Scatter Plots

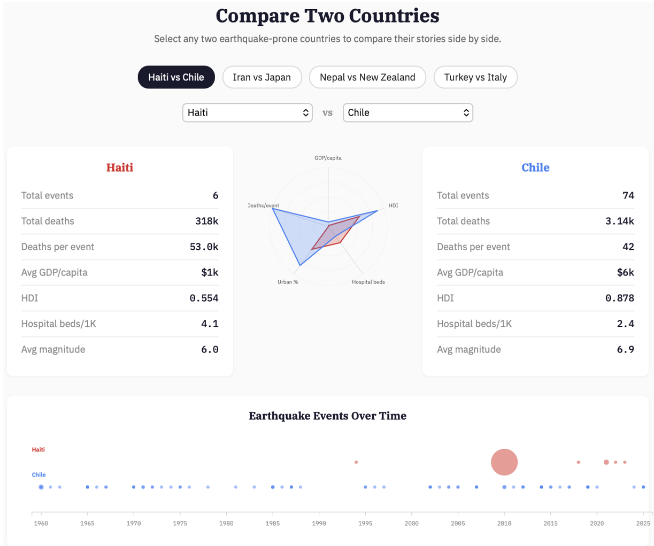
A 3×2 grid of scatter plots, each pairing one socio-economic indicator (GDP, HDI, hospital beds, urbanisation, population density, average magnitude) with deaths per event and labelled with its Pearson r. A brush on any panel cross-highlights the same countries in all six, letting the reader spot who is consistently vulnerable across axes.



Tools: D3 linked brushes, Pearson r in JS.
Lectures:
Data and Tabular data: Showing bipartite relationships and Pearson correlations
Interactions and More Interactive D3: D3 Brushing d3.brush, linked highlighting across multiple charts

2.6 V6 - Country Comparison Tool

The only fully free-exploration section. Two dropdowns (plus one-click suggestion chips for canonical pairs like Haiti-Chile and Iran-Japan) drive three linked views: stat cards, a radar chart overlaying normalised indicators for both countries, and a per-country event timeline.



Tools: D3 radial paths, HTML `<select>` with live filtering.
Lectures: *Interactions* (details on demand); *Designing Viz*; *Marks & Channels*.

3 Tools Summary

D3.js v7 powers every chart (scales, axes, transitions, geo projections, brushes, drag, animations). **TopoJSON** supplies world boundaries for the globe. **Scrollama** drives scroll-triggered steps in V2-V4. **Vite** provides the dev server and production bundling. **HTML/CSS** handles layout, sticky nav, and progress bar. **Python/pandas** pre-merges the NOAA, World Bank, and UNDP tables into the JSON files consumed by the site.

4 Implementation Breakdown

4.1 Core Visualisation (Minimum Viable Product)

These visualizations form the spine of the story, moving the reader directly from the initial anecdote to the statistical proof of the wealth-mortality gap:

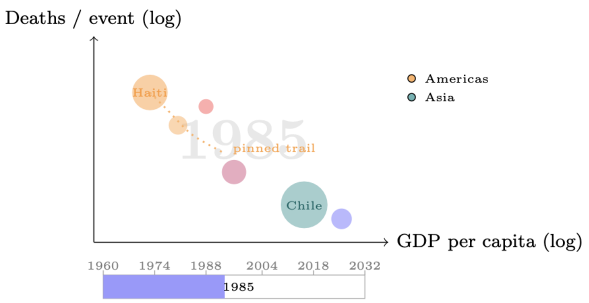
1. **The Hook** (V1) - Essential for setting up the central conflict (the Haiti vs. Chile anecdote).
2. **Magnitude Myth** (V3) - The first piece of structural evidence, proving that wealth, not just magnitude, dictates mortality on a global scale.
3. **Follow the Profile** (V4) - Expands the core argument to show how multidimensional socioeconomic factors align with death tolls.
4. **Correlation scatter plots** (V5) - The definitive statistical evidence, providing the mathematical proof (correlations) of the story's thesis.

5. **The Scrollama-driven narrative** that connects them.

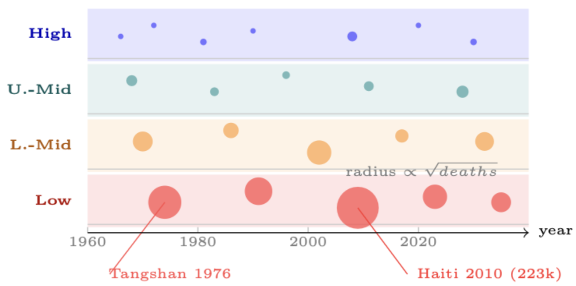
4.2 Enhancements (Can Be Dropped Without Losing Core Message)

These visualizations enhance the experience but are not strictly required to prove the central thesis:

1. **Interactive globe** (V2) - Optional context. It establishes where earthquakes happen globally, but the main story is about why they kill, which relies on socioeconomic data rather than geography
2. **Country comparison tool** (V6) - The narrative argument is already proven by the time the reader reaches this section; this tool simply allows them to interactively explore the data on their own terms.
3. **Gapminder animation** (To be implemented in the website) - GDP per capita vs. deaths per event, animated across decades



4. **Income-group timeline** (To be implemented in the website) - Four horizontal bands (High / Upper-Middle / Lower-Middle / Low income). Each event is a circle at its year with vertical jitter, sized by deaths and coloured by income group.



4.3 Data Pipeline

Raw inputs are the NOAA Significant Earthquake Database (CSV, one row per event) and the World Bank + UNDP indicator tables (wide-format CSVs, one row per country-year). A pandas script in the repository joins events to country-year indicators on `(country, year)`, derives per-event fields (deaths, magnitude, GDP at event time, income group), and writes five compact JSON artefacts that the site consumes: `earthquakes.json` (events), `countries.json` (country metadata), `country_year.json`, `paired_events.json` (country-pair data for V4), and `world-110m.json` (TopoJSON for V2).

4.4 Narrative Arc

The sections are ordered carefully to move the reader from a dramatic anecdote to statistical evidence, and finally to free exploration. Here is how each visualization builds the narrative:

- **The Hook**: Starts with a powerful anecdote (Haiti 2010 vs. Chile 2010). It anchors the data in a human reality, posing the central question of why outcomes are so drastically unequal despite similar (or stronger) seismic forces.

- **Interactive Globe (Enhancement)**: Transitions from the anecdote to the global scale. It shows that while seismic events are largely tied to geography (like the Ring of Fire), the resulting death tolls are not, setting up the need to look beyond magnitude.

- **Magnitude Myth**: Strips away geography to focus on a single axis: average deaths per event. By coloring dots by income group and sizing them by magnitude, it clearly reveals the wealth-mortality gap, showing that geology (size) is scattered, but wealth (color) dictates survival.

- **Follow the Profile**: Expands the narrative from a single axis to multiple dimensions. It threads the same countries through socioeconomic indicators (GDP, HDI, Hospital Beds) allowing the reader to trace how countries with different socioeconomic profiles (e.g., high-income vs. low-income) progress across these variables and how those differences relate to death outcomes

- **Correlation Scatter Plots**: Provides the mathematical evidence. It breaks down the multiple dimensions into individual scatter plots, displaying the strong negative correlations between national wealth/infrastructure metrics and earthquake mortality.

- **Comparison Tool (Enhancement)**: Serves as an open-ended conclusion. Having guided the reader through the evidence, this tool lets them explore the data on their own terms by putting any two earthquake-prone countries head-to-head across all metrics and historic events.